

Hierarchical Bayesian Exit Velocity Talent Model

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September 23, 2025

1 Task

My task is to build a model that estimates a hitter's underlying exit velocity ability when facing MLB-level competition. Using observed batted ball outcomes found in exit velo project data csv, project each hitter's expected average exit velocity into the 2024 season. For example: based on prior observations, you might predict that a given batter will have an average exit velocity on batted balls in play of 89 MPH when playing in the MLB in 2024. The submission should consist of two components:

- A written report in PDF format. This report should contain: A clear explanation of your data preparation process. A thorough description of your modeling process and methodology.
- Results of your model, discussing strengths, limitations, and future work.

2 Overview

2.1 Introduction

This analysis implements and justifies an ARIMAX (AutoRegressive Integrated Moving Average with exogenous variables) model for forecasting batter exit velocity. The absolute value of exit velocity will be used as a proxy to for exit velocity talent to simplify the formulation of the problem at this point.

We have data from 3,715 batters across MLB, AAA, and AA levels to generate 2024 performance projections. A total of 1344651 observations spanning 4 seasons from 2019 to 2023.

For feature selection, I have explored the given variables, categorical and continuous and I performed data engineering to build additional ones, for example, taking advantage that both launch angle and spray angle variables showed a visual sweet spot with higher exit velocity, the idea is transforming such variables to classify sweet spot and later, build statistics at the player level eg percentage of hits that are at the sweet spot.

Other variables such as batter age, batter height, or -same hand used eg both pitch and batter used same hand- were also relevant to hypothesize, nevertheless I did not find enough evidence that those were relevant to characterize exit velocity and then I did not include them in the model formulation at this time. I found that hang time was also relevant for when home run for example, the homerun observations were nested on high values of average hang time per player (4-6 sec) but it did not directly associated with exit velocity.

Future opportunities can be on exploring different structures of the model approach to incorporate such information.

2.2 Data Exploratory Analysis

I analyzed the data at the event-level and batter-level of granularity, in order to undertend the nested associations in data and the correlations between variables as a first assessment of portential predictors.

2.2.1 Exit Velocity Distribution

The mean exit velocity from empirical distribution is 88.00 mph (sd 14.48 mph), min = 2.6, max = 145.31. The MLB league showed an slightly higher mean exit velocity when compared with AAA, AA. Repsectively mean and sd, AA: (87.62,14.19), AAA: (87.82,14.32), MLB: (88.38,14.78) mph.

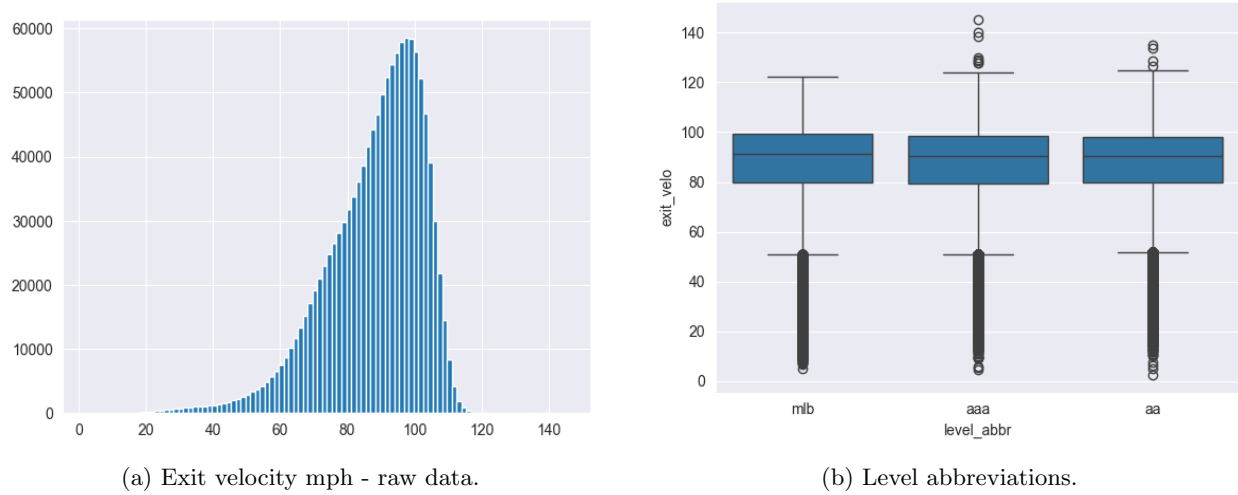


Figure 1: Comparison of results across two plots.

The exit velo absolute value is a proxy in this case for exit velo talent assessment, given that the home run shows the distribution of highest mean exit velo when compared with other outcomes.

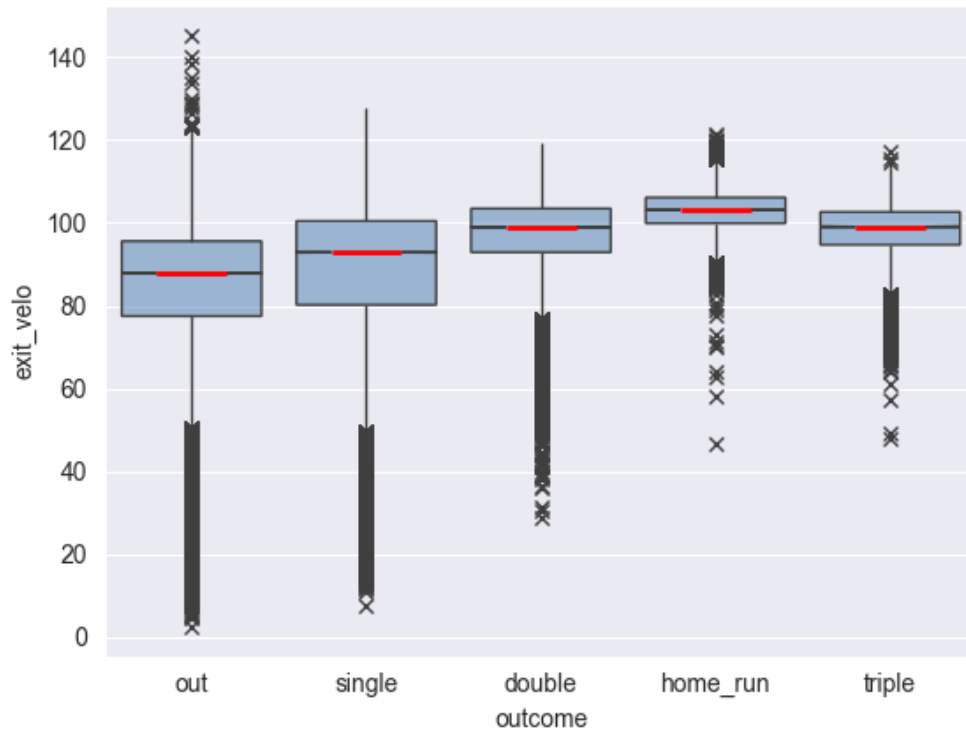


Figure 2: Outcomes.

2.2.2 Correlations

Exit velo shows correlation with launch and spray angle, being positive. The Launch angle is highly correlated with hangtime, we wont explore hang time at this point.

Other variables player intrinsic such as age and height wont be evaluated at this time, given the small correlation but opportunities in the future to evaluate in a second formulation of the model.

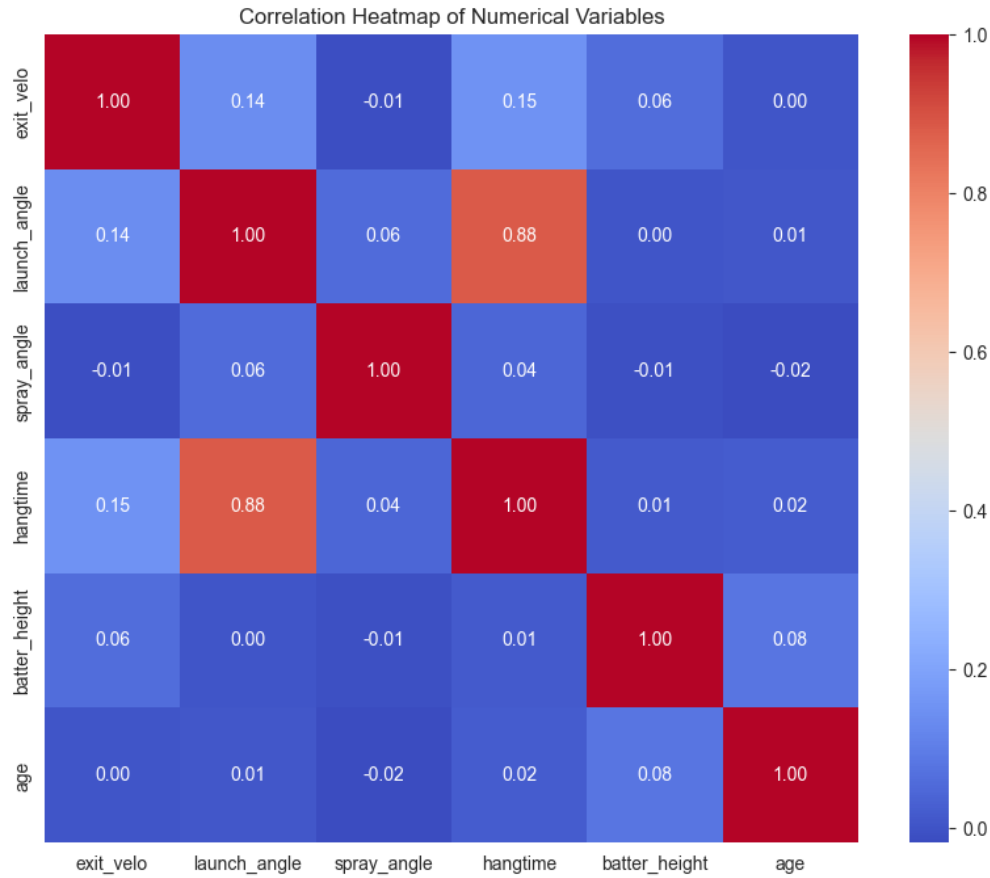


Figure 3: Correlations.

2.2.3 Pitch data

The pitch information is also relevant for the model assessment in the future. From the data exploration, the group FB is associated with the highest mean exit velo whenc ompared with other pitch groups. Optimal strategies for pitch type/group will be important for OS and BB types for example.

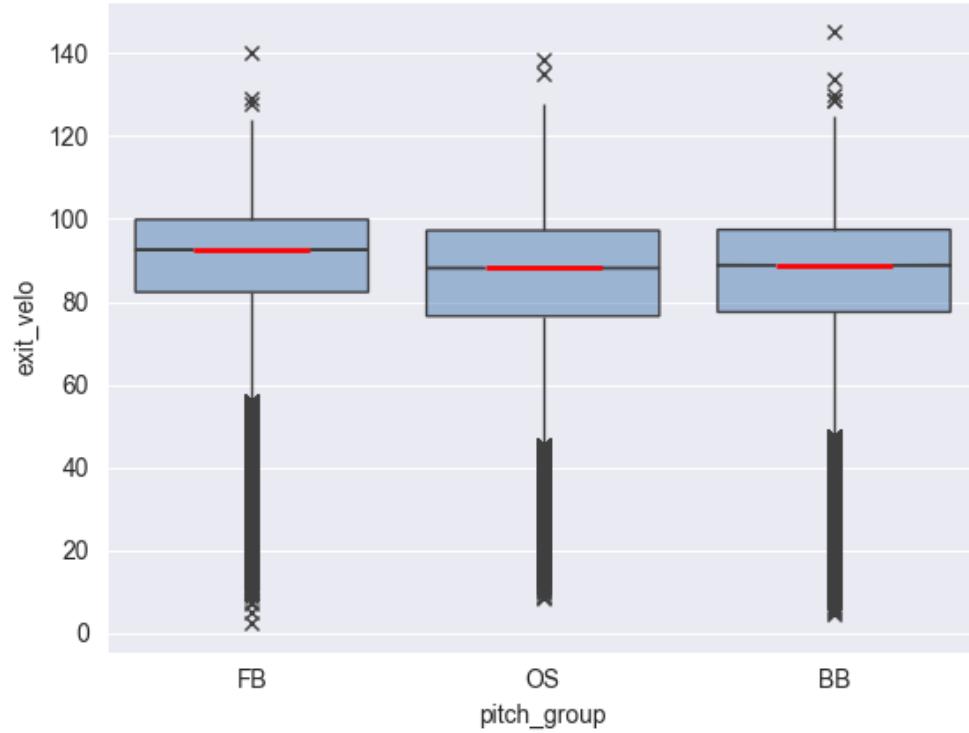


Figure 4: pitch group.

The hand type match was not relevant for modeling purposes.

2.2.4 Batter age and height data

The batter information is also relevant for the model assessment in the future. From the data exploration, those does not show a clear pattern for exit velocity. Strictly, there is a sweet spot for height between , that relationship can be evaluated in the future.



Figure 5: batter age by seasons.

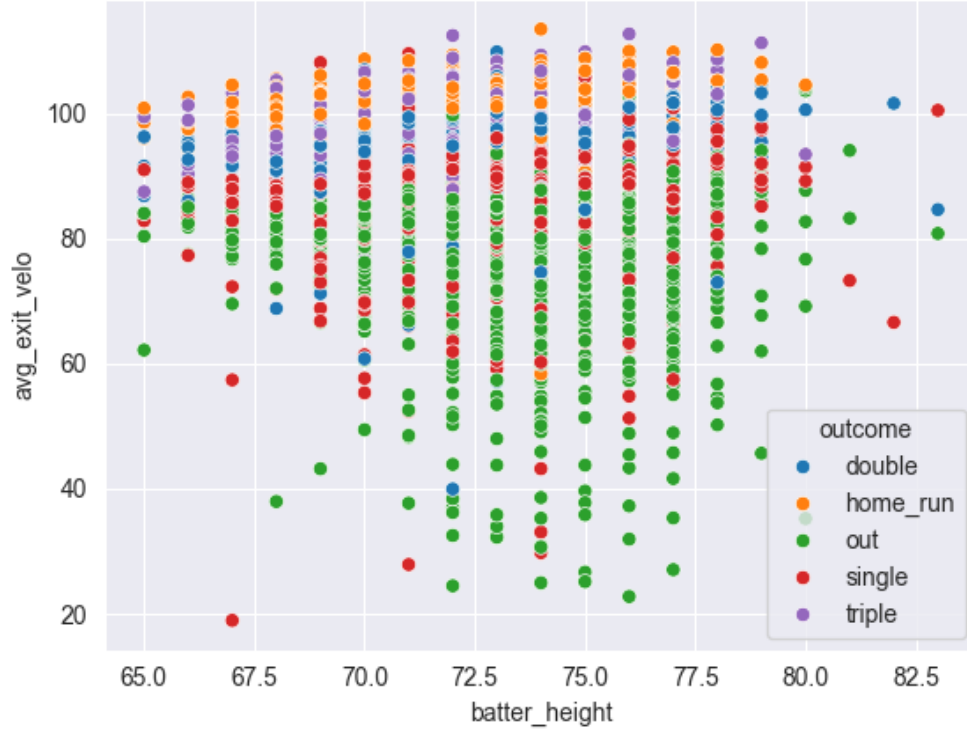
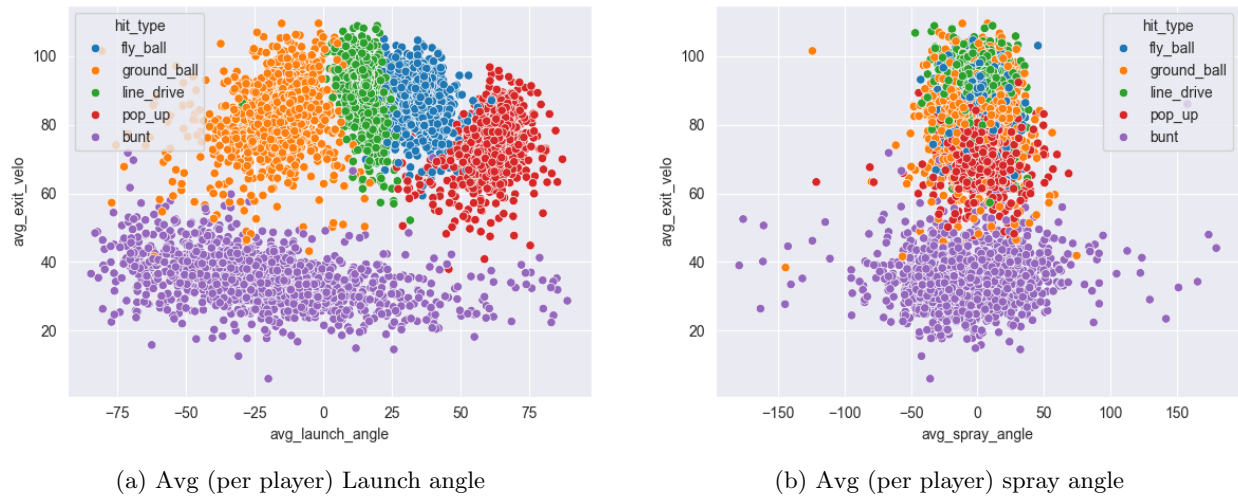


Figure 6: batter height.

2.2.5 Player level of granularity, the sweet spot: spray and launch angle

To be fair, the sweet spot in launch angle depends on the hit type. For simplicity, I defined the sweet spot in the overall, first defining the exit velo threshold on the 90 percentile. Later the range of spray and launch angle are as per percentiles 25 to 75 of such data.

It is important to assess the frequency the player hits the sweet spot.



(a) Avg (per player) Launch angle

(b) Avg (per player) spray angle

Figure 7: Sweet spot.

When the player hits the sweet spot, the exit velo distribution is as follows.

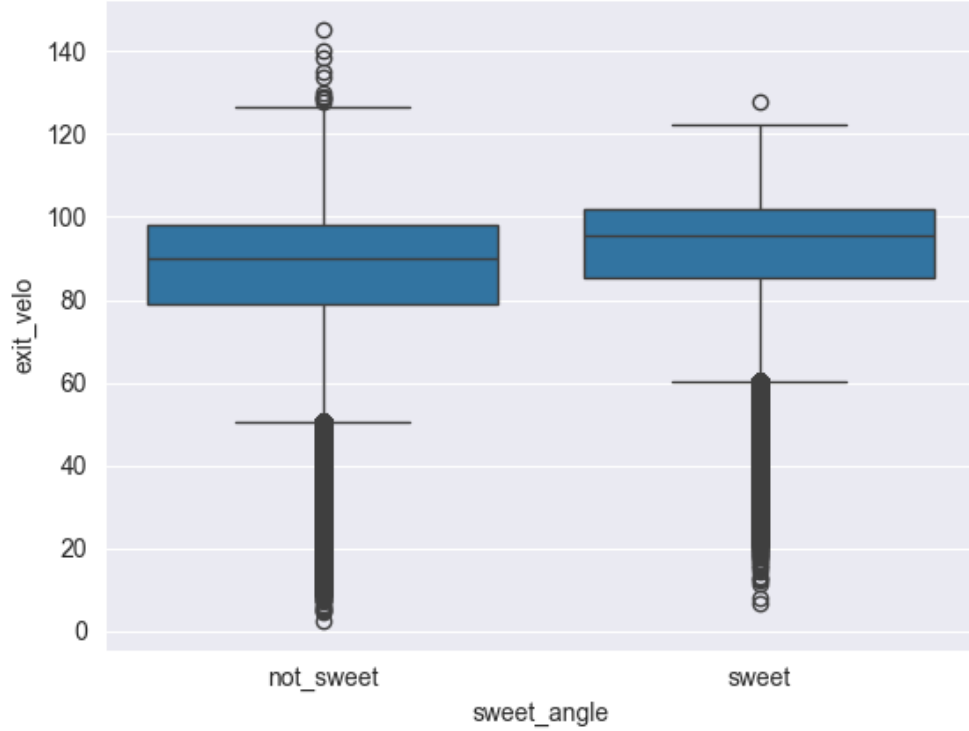


Figure 8: Sweet spot.

3 Mathematical Formulation

3.1 Per-Batter Forecasting with Auto-ARIMAX Models

I employ a hierarchical ARIMAX (AutoRegressive Integrated Moving Average with eXogenous variables) framework to forecast individual player exit velocity performance in Major League Baseball for the 2024 season. The model accounts for player development trajectories, league transitions, and batting mechanics evolution.

3.2 Notation

Let $y_{i,t}$ denote the average exit velocity for player i in season t , where $i \in \{1, 2, \dots, N\}$ indexes players and $t \in \{2019, 2020, 2021, 2022, 2023\}$ represents the historical seasons. Our objective is to forecast $\hat{y}_{i,2024}$ for each eligible player.

3.3 Feature Engineering

3.3.1 League Progression Features

For each player-season combination, I construct league share variables:

$$s_{i,t}^{(l)} = \frac{n_{i,t}^{(l)}}{\sum_{l' \in \mathcal{L}} n_{i,t}^{(l')}} \quad l \in \mathcal{L} = \{\text{AA}, \text{AAA}, \text{MLB}\} \quad (1)$$

where $n_{i,t}^{(l)}$ represents the number of plate appearances for player i in season t at league level l .

To avoid perfect multicollinearity ($\sum_{l \in \mathcal{L}} s_{i,t}^{(l)} = 1$), I employ $(|\mathcal{L}| - 1)$ dummy encoding, using MLB as the reference category:

$$\mathbf{S}_{i,t} = [s_{i,t}^{(\text{AA})}, s_{i,t}^{(\text{AAA})}]^T \quad (2)$$

3.3.2 Batting Mechanics Features

I extract season-level angle-based features from batted ball events, in particular, in the exploratory notebook I obtained the ranges of spray and launch angle eg percentile 25 and 75 of each, that reaches the percentile 90 of the exit velo. For launch angle α and spray angle β :

sweet Spot Rate:

$$\text{sweetRate}_{i,t} = \frac{1}{|B_{i,t}|} \sum_{b \in B_{i,t}} \mathbb{I}(\alpha_b \in [2.25, 21.9] \cap \beta_b \in [-2.7, 12.9]) \quad (3)$$

where $B_{i,t}$ is the set of batted ball events for player i in season t .

Launch Angle Dispersion:

$$\text{LaunchDist}_{i,t} = \frac{1}{|B_{i,t}|} \sum_{b \in B_{i,t}} (\alpha_b - 12.0)^2 \quad (4)$$

Spray Angle Consistency: Using circular statistics for spray angle β :

$$\text{SprayCos}_{i,t} = \frac{1}{|B_{i,t}|} \sum_{b \in B_{i,t}} \cos(\beta_b) \quad (5)$$

$$\text{SpraySin}_{i,t} = \frac{1}{|B_{i,t}|} \sum_{b \in B_{i,t}} \sin(\beta_b) \quad (6)$$

$$\text{SprayDev}_{i,t} = \frac{1}{|B_{i,t}|} \sum_{b \in B_{i,t}} |\beta_b| \quad (7)$$

3.3.3 Hit Type Distribution

I model the evolution of batted ball outcome distributions using one-hot encoding:

$$\mathbf{H}_{i,t} = [\text{Ground Ball}_{i,t}, \text{Line Drive}_{i,t}, \text{Fly Ball}_{i,t}, \text{Pop-up}_{i,t}]^T \quad (8)$$

where each component represents the seasonal proportion of that hit type.

3.4 ARIMAX Model Specification

3.4.1 Exogenous Feature Matrix

The complete exogenous feature matrix for player i at time t is:

$$\mathbf{X}_{i,t} = [\mathbf{S}_{i,t}^T, \mathbf{A}_{i,t}^T, \mathbf{H}_{i,t}^T]^T \quad (9)$$

where $\mathbf{A}_{i,t} = [\text{sweetRate}_{i,t}, \text{LaunchDist}_{i,t}, \text{SprayCos}_{i,t}, \text{SpraySin}_{i,t}, \text{SprayDev}_{i,t}]^T$.

3.4.2 Multicollinearity Handling

To prevent estimation issues, I remove constant columns and address perfect linear dependencies:

$$\mathbf{X}_{i,t}^{\text{clean}} = \text{DropConstant}(\mathbf{X}_{i,t}) \quad (10)$$

$$\text{where } \text{DropConstant}(\mathbf{X}) = \{x_j \in \mathbf{X} : \text{Var}(x_j) > \epsilon\} \quad (11)$$

with tolerance $\epsilon = 10^{-12}$.

3.4.3 ARIMAX Formulation

For each player i , I estimate an individual ARIMAX(p_i, d_i, q_i) model:

$$(1 - \phi_{i,1}L - \dots - \phi_{i,p_i}L^{p_i})(1 - L)^{d_i}(y_{i,t} - \mathbf{X}_{i,t}^T\boldsymbol{\beta}_i) = (1 + \theta_{i,1}L + \dots + \theta_{i,q_i}L^{q_i})\epsilon_{i,t} \quad (12)$$

where:

- L is the lag operator: $L^k y_t = y_{t-k}$
- $\phi_{i,j}$ are autoregressive parameters
- $\theta_{i,j}$ are moving average parameters
- $\boldsymbol{\beta}_i$ is the exogenous variable coefficient vector
- $\epsilon_{i,t} \sim \mathcal{N}(0, \sigma_i^2)$ are white noise innovations

3.5 Model Selection

3.5.1 Automated Order Selection

I employ the auto-ARIMA algorithm with stepwise search to determine optimal orders (p_i, d_i, q_i) :

$$(p_i^*, d_i^*, q_i^*) = \underset{(p,d,q) \in \mathcal{S}}{\text{argmin}} \text{AIC}_i(p, d, q) \quad (13)$$

where $\mathcal{S} = \{(p, d, q) : p, q \in \{0, 1, 2, 3\}, d \in \{0, 1, 2\}\}$ and:

$$\text{AIC}_i(p, d, q) = -2 \log L_i(p, d, q) + 2k_i \quad (14)$$

with L_i being the likelihood function and k_i the number of parameters.

3.5.2 Cross-Validation

When sufficient data permits ($T_i \geq 4$), I employ rolling origin cross-validation:

$$\text{RMSE}_{\text{CV},i} = \sqrt{\frac{1}{T_i - 3} \sum_{t=3}^{T_i-1} (y_{i,t+1} - \hat{y}_{i,t+1|t})^2} \quad (15)$$

where $\hat{y}_{i,t+1|t}$ is the one-step-ahead forecast using data through time t .

3.6 Forecasting

3.6.1 Exogenous Variable Projection

For 2024 forecasting, I project exogenous variables using recent trends. For angle-based features:

$$\hat{x}_{i,2024} = x_{i,2023} + \frac{1}{2} \sum_{j=1}^2 (x_{i,2023-j+1} - x_{i,2023-j}) \quad (16)$$

For league indicators, I set $\mathbf{S}_{i,2024}$ based on target league level.

3.6.2 Point and Interval Forecasts

The 2024 point forecast for player i is:

$$\hat{y}_{i,2024} = \mathbb{E}[y_{i,2024} | \mathcal{F}_{2023}, \mathbf{X}_{i,2024}] \quad (17)$$

Prediction intervals are constructed using the forecast error variance:

$$\text{PI}_{1-\alpha} = \hat{y}_{i,2024} \pm z_{\alpha/2} \sqrt{\text{Var}(\hat{y}_{i,2024} | \mathcal{F}_{2023})} \quad (18)$$

3.7 Fallback Methodology

For players where ARIMAX estimation fails, I employ a level-adjusted persistence model:

$$\hat{y}_{i,2024}^{\text{fallback}} = y_{i,2023} + \Delta_{\text{level}} \quad (19)$$

where $\Delta_{\text{level}} \in \{-4.5, -2.5, 0\}$ represents the expected performance differential between league levels, with uncertainty estimated from historical differences:

$$\sigma_{\text{fallback},i}^2 = \frac{1}{T_i - 1} \sum_{t=2}^{T_i} (y_{i,t} - y_{i,t-1})^2 \quad (20)$$

3.8 Model Justification

This ARIMAX procedure is justified for baseball forecasting because of the temporal dependencies and because of

1. **Player Development:** The autoregressive components capture skill development momentum and learning curves inherent in athletic performance.
2. **Uncertainty Quantification:** This probabilistic framework provides prediction intervals essential for risk assessment in player evaluation contexts.

The model balances complexity with parsimony, incorporating domain knowledge while maintaining sufficient flexibility to capture individual development trajectories in professional baseball. By using automatic order selection, the framework scales across hundreds of heterogeneous batters without requiring manual tuning.

3.9 Model Result

The ARIMA model demonstrated a median RMSE of 3.0 and MAE of 2.6, indicating that for most players the model predictions were within 3–4 units of the observed values. However, the error distribution is highly skewed, with a small number of extreme outliers (RMSE above 100), suggesting that ARIMA struggles with certain player trajectories. AIC values were generally negative (mean = -43, min = -104), which indicates reasonable model fit overall, but the wide range (up to 14) highlights that ARIMA is not uniformly optimal across all cases.

Table 1: Validation metrics summary.

Statistic	val_rmse	val_mae	val_aic
Mean	5.28	4.47	-43.25
Std	9.81	7.65	24.64
Min	0.11	0.11	-104.40
25%	1.91	1.62	-47.56
50%	3.01	2.56	-38.70
75%	5.22	4.69	-36.03
Max	126.87	99.22	13.92

The pitch information is also relevant for the model assessment in the future. From the data exploration, the group FB is associated with the highest mean exit velo whenc ompared with other pitch groups. Optimal strategies for pitch type/group will be important for OS and BB types for example.

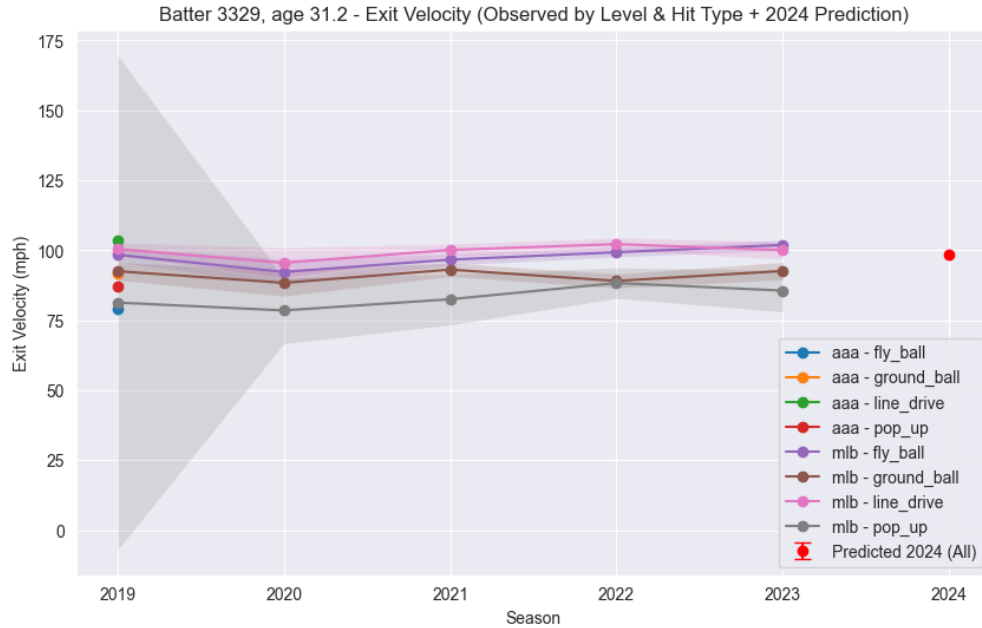


Figure 9: batter example.

3.10 References and Notes

Github Repository:

<https://github.com/mariaob1201/Miami-marlins-challenge.git>

I employed autoarima from pdrima to systematically identify ARIMA hyperparameters for each batter trajectory, replacing grid search parameter optimization. The presence of large outliers indicates limitations of the model's flexibility for all players.

Future Directions

Bayesian model

In addition to the ARIMA model, a Bayesian model was explored on a small sample of the total data using PyMC, formulating a hierarchical structure based on the data analysis. The results are not reported, but the corresponding code is included in the repository.

Level Equivalencies per Batter

To address the research question about level equivalencies per batter, one approach was to assign an ELO score to compare the average exit velocity across scenarios, using pairwise comparisons of players by batter ID. The resulting score provides a relative level for each player when compared with the others.

References

- <https://pypi.org/project/pmdarima/>
- <https://github.com/alkaline-ml/pmdarima>
- <https://yougoprobaseball.com/hitting-mechanics-swing/>
- <https://github.com/mariaob1201/time-series-demo>
- <https://github.com/mariaob1201/baseball-phillies-hackathon>

3.11 Derivables

The analysis was structured into three main components, each implemented in separate scripts or notebooks:

- **data_exploration.ipynb**: Provides the initial assessment of the dataset, including an overview of variable distributions and preliminary descriptive statistics.
- **arima.py**: Implements the ARIMA methodology and overall workflow for model fitting, data preprocessing, and feature engineering.
- **rev.ipynb**: Integrates and visualizes the results produced by **arima.py**, and merges them with the evaluation dataset for downstream comparison and validation.
- **miami_marlins.csv**: is the final report.
- **forecasting2024_mlb_maria.csv**: is the player level forecasting for 2024.

I used chat gpt to order my code a little bit, so it can be easy to evaluate, also to fix errors in latex.